The thallus is made up of branched structures called hyphae. A large number of hyphae constitute the mycelium. Three types of mycelia are seen namely primary mycelium, secondary mycelium and tertiary mycelium. The primary mycelium develops from the germination of basidiospore. It is septate, uninucleate and haploid. It is also called **monokaryotic mycelium**. Fusion of two primary mycelium of opposite strains give rise to secondary mycelium or **dikaryotic mycelium**. The dikaryotic mycelium develops into hyphal cords called **Rhizomorphs**, and perennates the soil for a long period. The tertiary mycelium found in the fruit body called **basidiocarp**. Each cell of the hyphae posssess a cell wall made up of chitin and cell organelles like mitochondria, golgibodies, endoplasmic reticulum etc., are also present.

Asexual reproduction

Agaricus produces chlamydospores during asexual reproduction. During favourable condition the chlamydospores germinate to produce mycelium.

Sexual reproduction

Agaricus reproduces by sexual method but sex organs are absent. Majority of the species are heterothallic. *Agaricus bisporus* is a homothallic species. The opposite strains of mycelium fuse (somatogamy) and results in the formation of dikaryotic or secondary mycelium. Karyogamy takes place in basidium and it is immediately followed by meiosis giving rise to four haploid basidiospores. The basidiospores are borne on sterigmata. The subterranean mycelial strands called rhizomorphs posssess dense knots of dikaryotic hyphae. These knots develop into basidiocarps.

Basidiocarp

The mature basidiocarp is umbrella shaped and is divided into 3 parts namely stipe, pileus and gill. The stipe is thick, fleshy and cylindrical in structure. The upper part of the stipe possess a membranous structure called **annulus**. The upper convex surface is called **pileus** which is white or cream in colour (Figure 1.24). The inner surface of pileus shows radially arranged **gills** or lamellae. The gills vary in length. On both the sides of the gills a fertile layer called **hymenium** is present. The stipe is hollow from the centre and the central part is made up of loosely arranged hyphae whereas the periphery is made up of compactly arranged hyphae forming **pseudoparenchymatous** tissue. The gill region is divided into 3 regions. The central part of gill between two hymenial layers is called **Trama** (Figure 1.25). The subhymenial layers have closely compact tissue . The **hymenium** is the fertile layer and possess club shaped basidia. The basidium is interspersed with sterile hyphae called paraphysis. Each basidium bears 4 basidiospores, of these two basidiospores belong to (+) strain and other two of them will be (–) strain. The basidiospores are borne on stalk like structures called **sterigmata**. The basidiospore on germination produces the haploid primary mycelium.

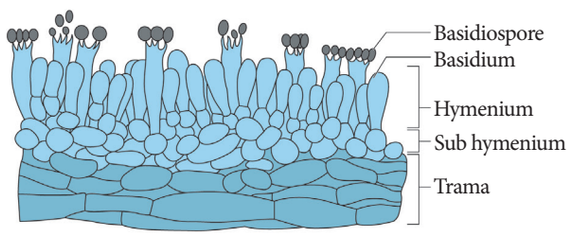


Figure 1.25: V.S. of *Agaricus* gill

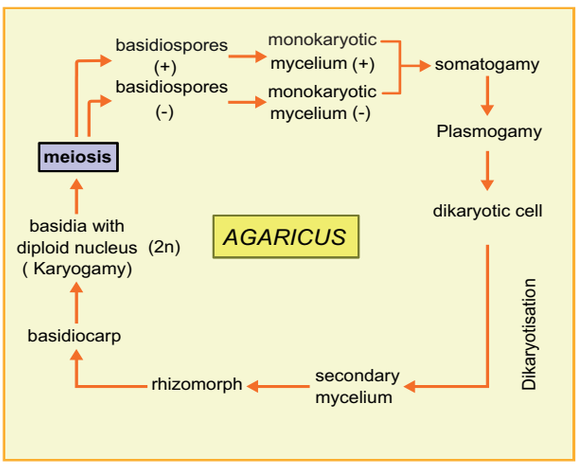


Figure 1.26: Life Cycle of *Agaricus*

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